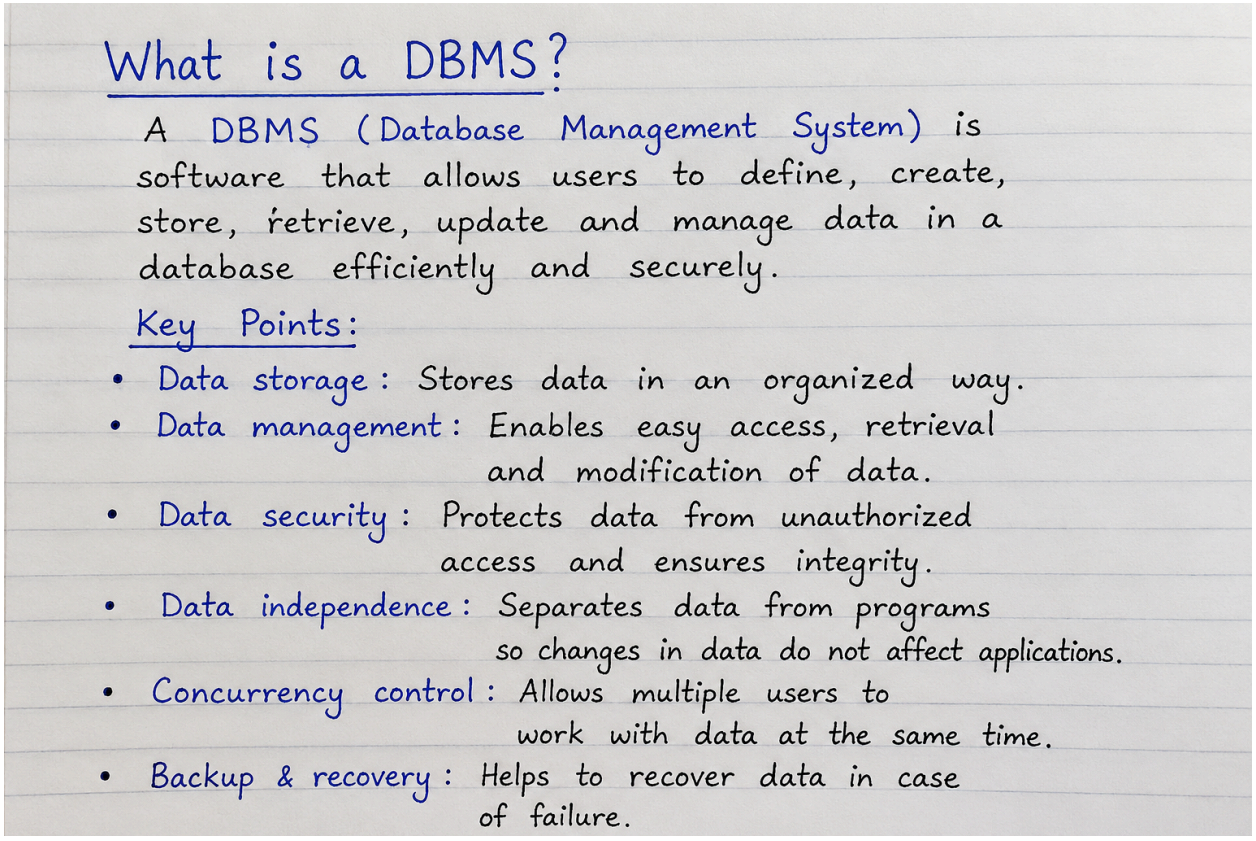
Week 1

What is a DBMS? Differentiate between DBMS and File System.



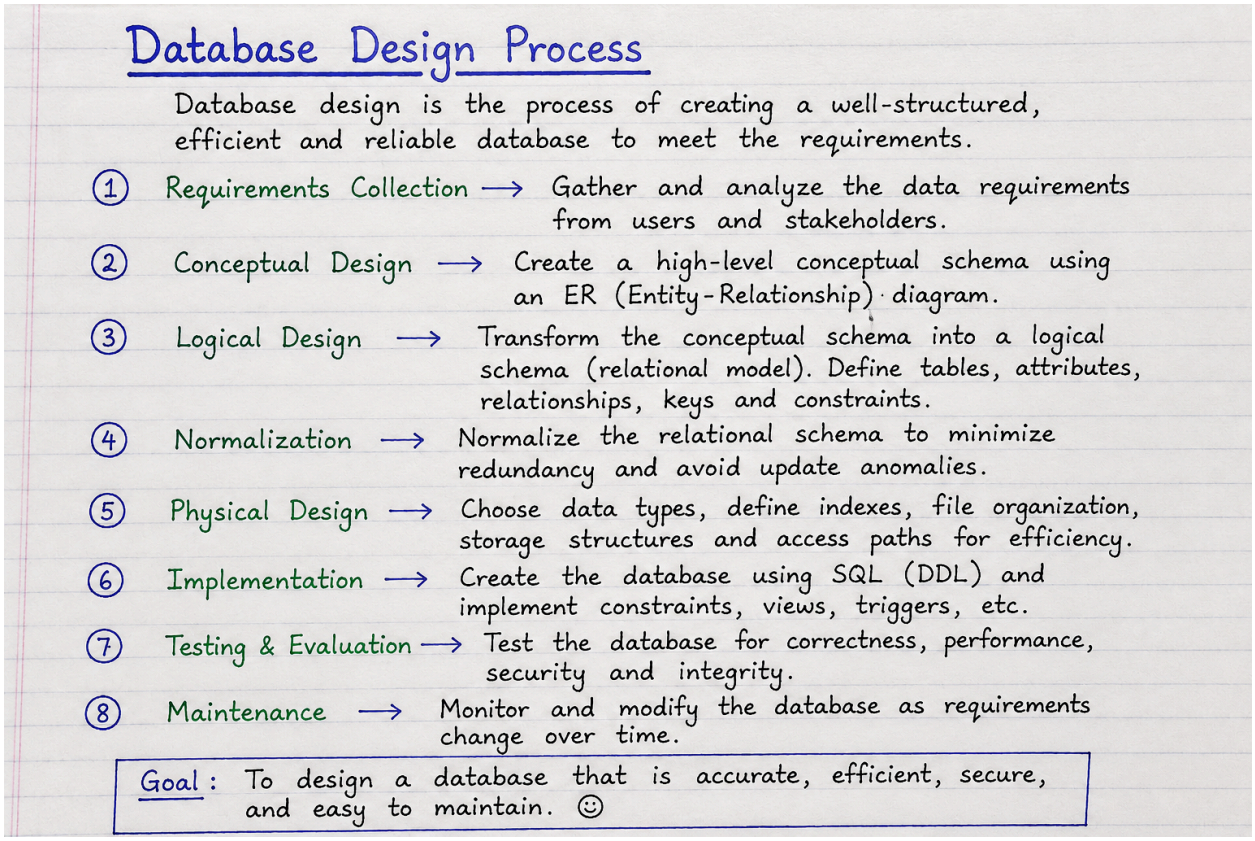
What is a DBMS?

A DBMS (Database Management System) is software that allows users to define, create, store, retrieve, update and manage data in a database efficiently and securely.

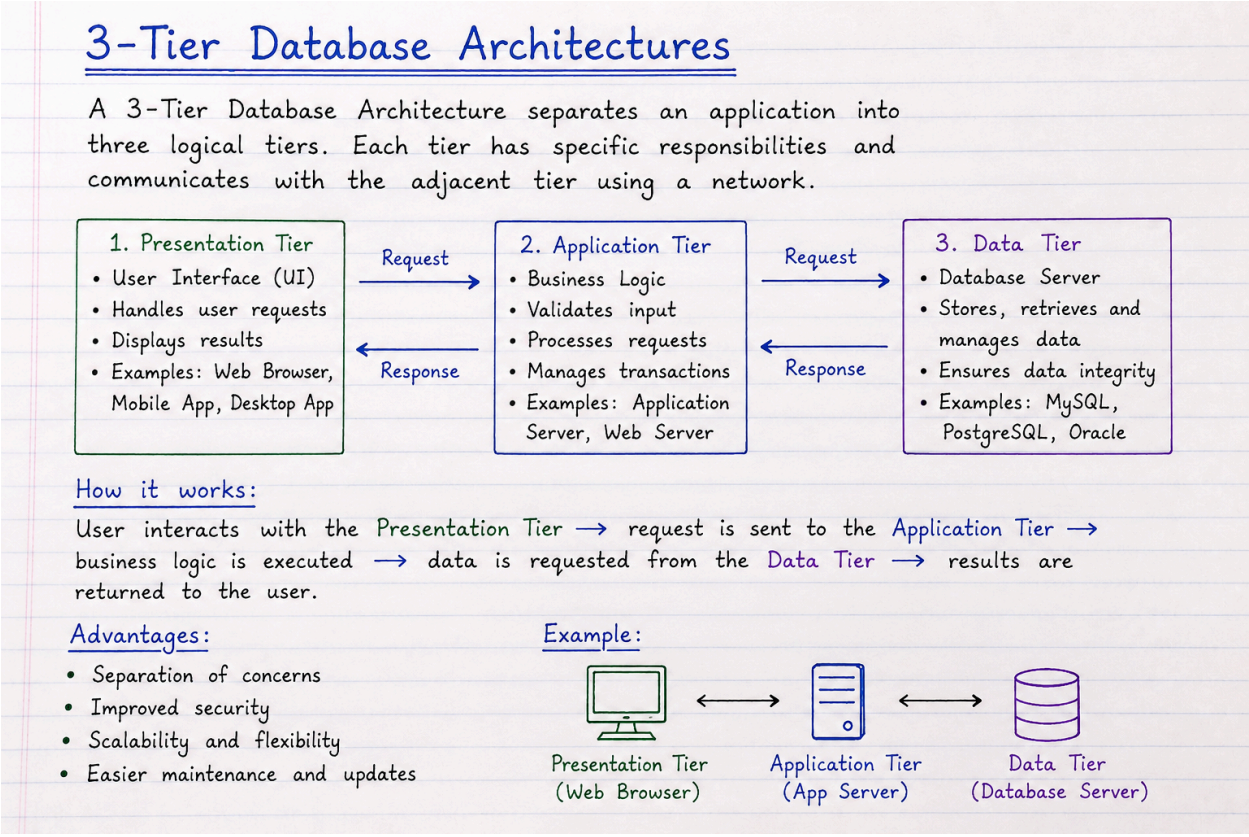
Key Points:

- **Data storage:** Stores data in an organized way.
- **Data management:** Enables easy access, retrieval and modification of data.
- **Data security:** Protects data from unauthorized access and ensures integrity.
- **Data independence:** Separates data from programs so changes in data do not affect applications.
- **Concurrency control:** Allows multiple users to work with data at the same time.
- **Backup & recovery:** Helps to recover data in case of failure.

Database Design Process

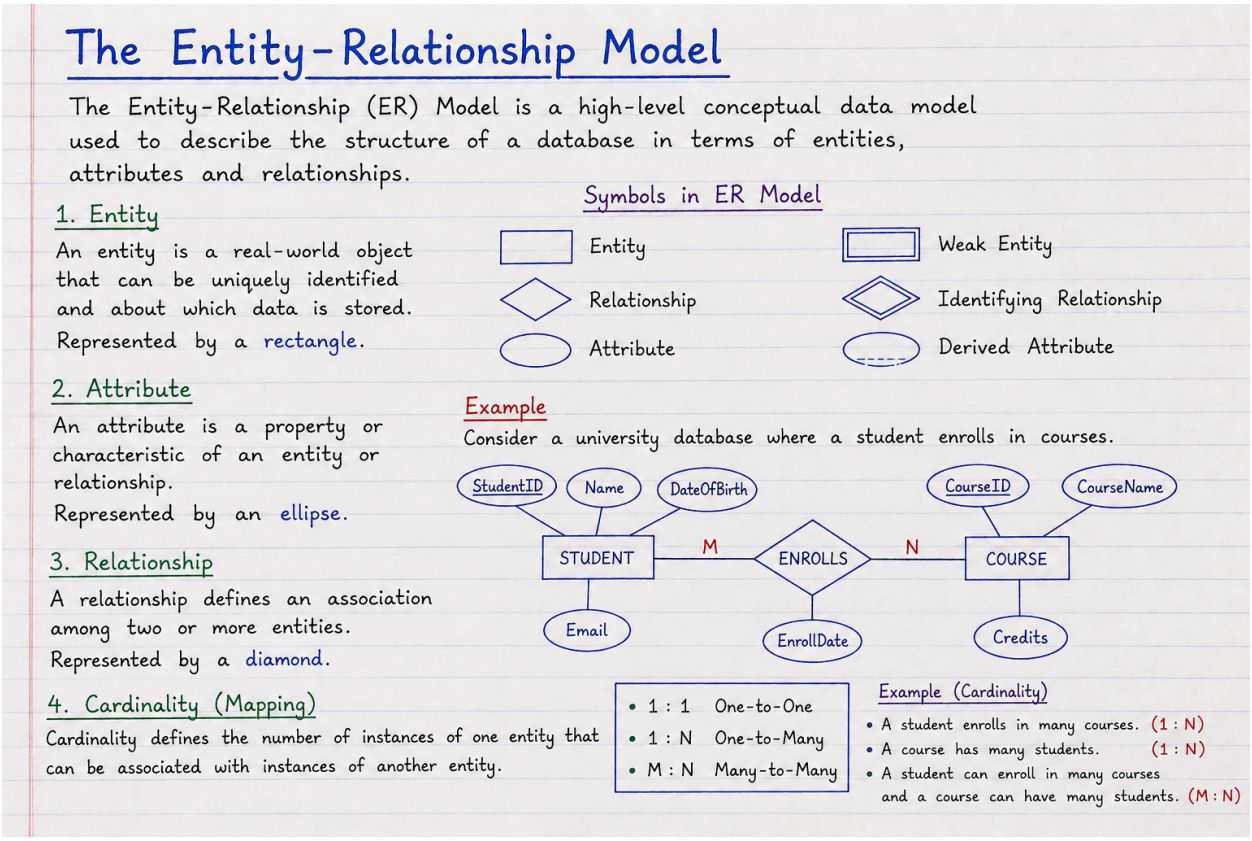


Database Architectures (3-Tier Database Architectures)



Week 2

The Entity–Relationship Model



Relationships & Cardinality

Relationships & Cardinality

Relationship: An association among two or more entities.
Cardinality: The number of instances of one entity that can be associated with instances of another entity through a relationship.

Types of Relationships & Cardinality

1. One-to-One (1 : 1)

One instance of an entity is associated with only one instance of another entity.

Example: A person has one passport.

---

2. One-to-Many (1 : N)

One instance of an entity is associated with many instances of another entity.

Example: A department has many employees.

---

3. Many-to-One (N : 1)

Many instances of an entity are associated with one instance of another entity.

Example: Many employees work in one department.

---

4. Many-to-Many (M : N)

Many instances of an entity are associated with many instances of another entity.

Example: Many students enroll in many courses.

Cardinality Notation

—— Single line : Partial participation (optional)

═══ Double line : Total participation (mandatory)

(0,1)	Minimum 0, Maximum 1	(optional one)
(1,1)	Minimum 1, Maximum 1	(exactly one)
(0,N)	Minimum 0, Maximum N	(zero or many)
(1,N)	Minimum 1, Maximum N	(one or many)

Key Points

- Cardinality helps define the nature of relationships.
- It is important for designing accurate ER diagrams and database schemas.

Keys — Super, Candidate, Primary, Alternate

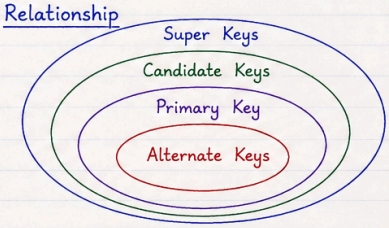
Keys — Super, Candidate, Primary, Alternate

- **Super Key :** A set of one or more attributes that can uniquely identify a tuple (row) in a relation.
- **Candidate Key :** A minimal super key. No proper subset of it can uniquely identify a tuple.
- **Primary Key :** The candidate key selected to uniquely identify tuples in a relation.
- **Alternate Key :** The candidate keys that are not chosen as the primary key.

Key Points

- Every table must have a primary key.
- A table can have one primary key.
- A table can have one or more candidate keys.
- Alternate keys = Candidate keys − Primary key.

Relationship



Example

Consider the relation STUDENT with the following attributes.

RollNo	StudentID	Email	Phone	Name	Dept
101	S001	s001@uni.edu	01710000001	Arafat	CSE
102	S002	s002@uni.edu	01710000002	Bithy	EEE
103	S003	s003@uni.edu	01710000003	Chayon	CSE
104	S004	s004@uni.edu	01710000004	Disha	BBA

Identify the keys:

→ RollNo alone can uniquely identify each student. So, {RollNo} is a candidate key.
→ StudentID alone can uniquely identify each student. So, {StudentID} is a candidate key.
→ Email alone can uniquely identify each student. So, {Email} is a candidate key.
→ Phone alone can uniquely identify each student. So, {Phone} is a candidate key.
→ Any combination of these (e.g., {RollNo, Name}, {StudentID, Dept}) is a super key, but not a candidate key (because they are not minimal).

Candidate Keys:
{RollNo}, {StudentID}, {Email}, {Phone}

Primary Key:
Let {RollNo} be chosen as the primary key.

Alternate Keys:
{StudentID}, {Email}, {Phone}

Super Keys (examples):
{RollNo}, {RollNo, Name}, {RollNo, Dept}, {StudentID, Email}, {Email, Phone}, etc.

ER Diagram Practice Examples

Scenario 1 — Training Institute Course Registration

Level: Easy **Concepts:** Entity sets · key attributes · M:N relationship · descriptive (relationship) attributes

The scenario

Bright Path Training Institute wants to computerise its registration desk. Every student is given a unique student ID, and the office also stores the student's name, e-mail address and phone number. Every course has a unique course code, a title, and a number of credit hours.

A student may register for as many courses as they wish, and the same course is normally taken by many students. Whenever a student registers for a course, the office records the semester in which the course was taken and the grade that was finally awarded.

A student who has not yet registered for anything is still kept in the system, and a newly announced course is stored even before anybody registers for it. Draw an ER diagram for this system.

Solution — entity sets and attributes

Entity set	Attributes (primary key underlined)
STUDENT	student_id, name, email, phone
COURSE	course_code, title, credit_hours

Solution — relationship sets

Relationship	Between	Ratio	Participation	Attributes
REGISTERS	STUDENT – COURSE	M : N	Partial on both sides	semester, grade

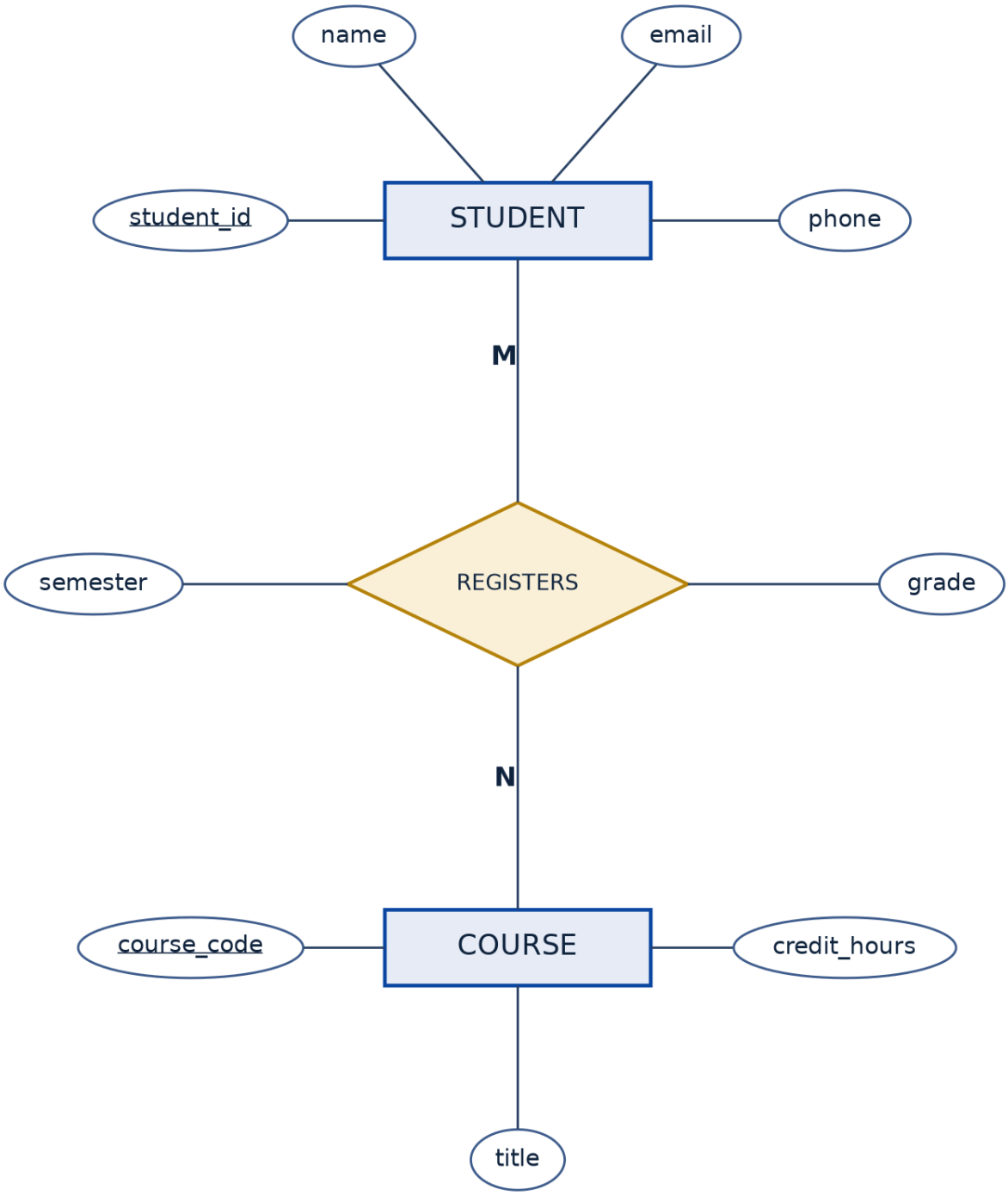


Figure 1 — ER diagram for Scenario 1: Training Institute Course Registration

Scenario 2 — City Public Library

Level: Easy **Concepts:** Two relationships on one entity · M:N · relationship attributes · total vs partial participation

The scenario

The City Public Library keeps a record of every book it owns. A book is identified by its ISBN, and the library also stores the title, edition and publisher. Each book is written by one or more authors; an author is identified by an author ID and has a name. An author may of course have written several books, and the library stores an author only if it owns at least one of that author's books.

Members are identified by a membership ID, and the library stores name, address and phone number. A member may borrow many books over a period of time, and the same book may be borrowed by many members at different times. For every borrowing the library notes the issue date, the due date and the return date.

A membership record is created only at the moment of the first borrowing, so every member in the database has borrowed at least one book. Draw an ER diagram.

Solution — entity sets and attributes

Entity set	Attributes (primary key underlined)
BOOK	ISBN, title, edition, publisher
MEMBER	member_id, name, address, phone
AUTHOR	author_id, author_name

Solution — relationship sets

Relationship	Between	Ratio	Participation	Attributes
BORROWS	BOOK – MEMBER	M : N	Total on MEMBER, partial on BOOK	issue_date, due_date, return_date
WRITTEN_BY	BOOK – AUTHOR	M : N	Total on AUTHOR, partial on BOOK	—

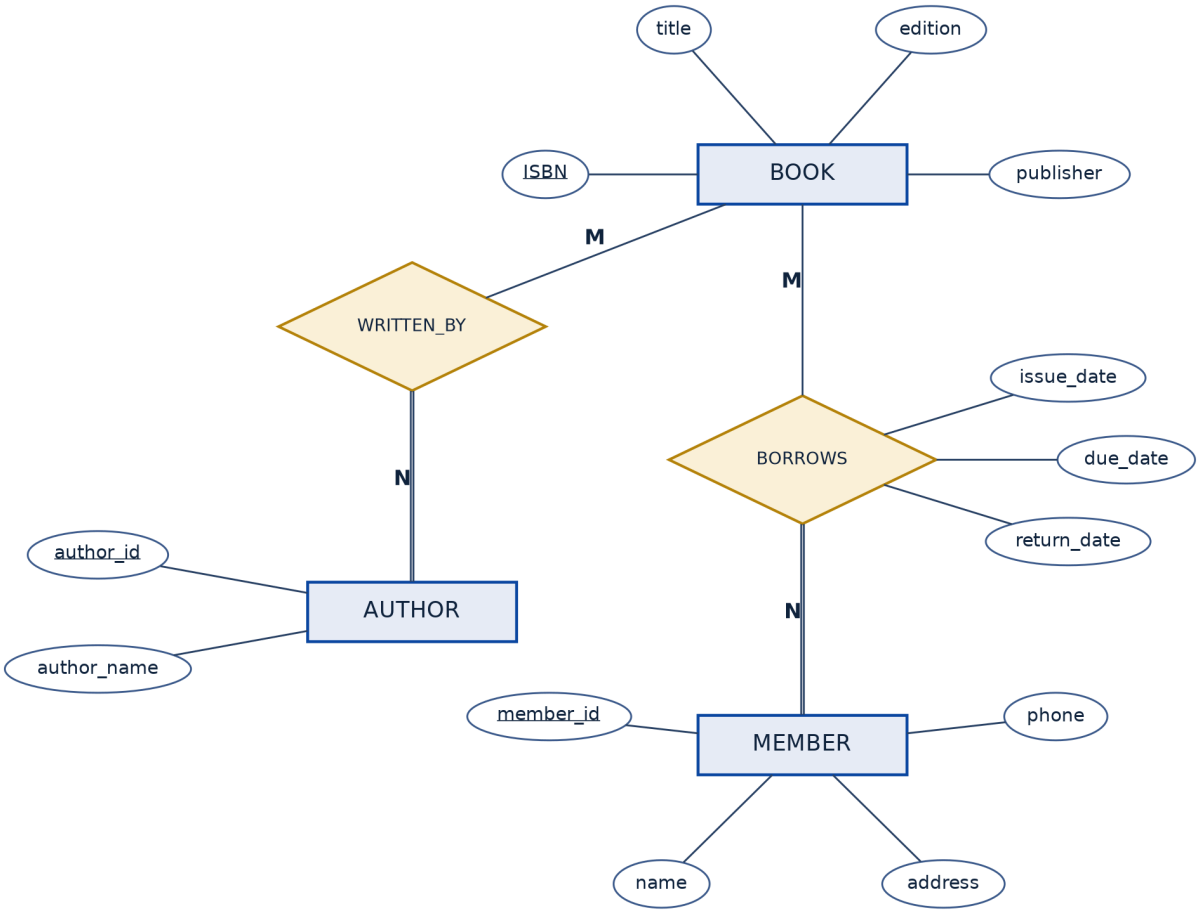


Figure 2 — ER diagram for Scenario 2: City Public Library

Scenario 3 — Company Employees and Departments

Level: Easy **Concepts:** 1:N and 1:1 in one diagram · total vs partial participation · one entity in two roles

The scenario

A manufacturing company stores, for every employee, an employee ID, name, designation and salary. For every department it stores a department number, department name and location.

Each employee works for exactly one department and must be assigned to one; a department may employ many people. Each department has exactly one manager, and that manager is one of the company's own employees. An employee may manage at most one department, and most employees manage none. The date on which a manager took charge of the department is recorded.

Draw an ER diagram showing the correct cardinality ratios and participation constraints.

Solution — entity sets and attributes

Entity set	Attributes (primary key underlined)
EMPLOYEE	emp_id, emp_name, designation, salary
DEPARTMENT	dept_no, dept_name, location

Solution — relationship sets

Relationship	Between	Ratio	Participation	Attributes
WORKS_FOR	EMPLOYEE – DEPARTMENT	N : 1	Total on EMPLOYEE, partial on DEPARTMENT	—
MANAGES	EMPLOYEE – DEPARTMENT	1 : 1	Total on DEPARTMENT, partial on EMPLOYEE	start_date

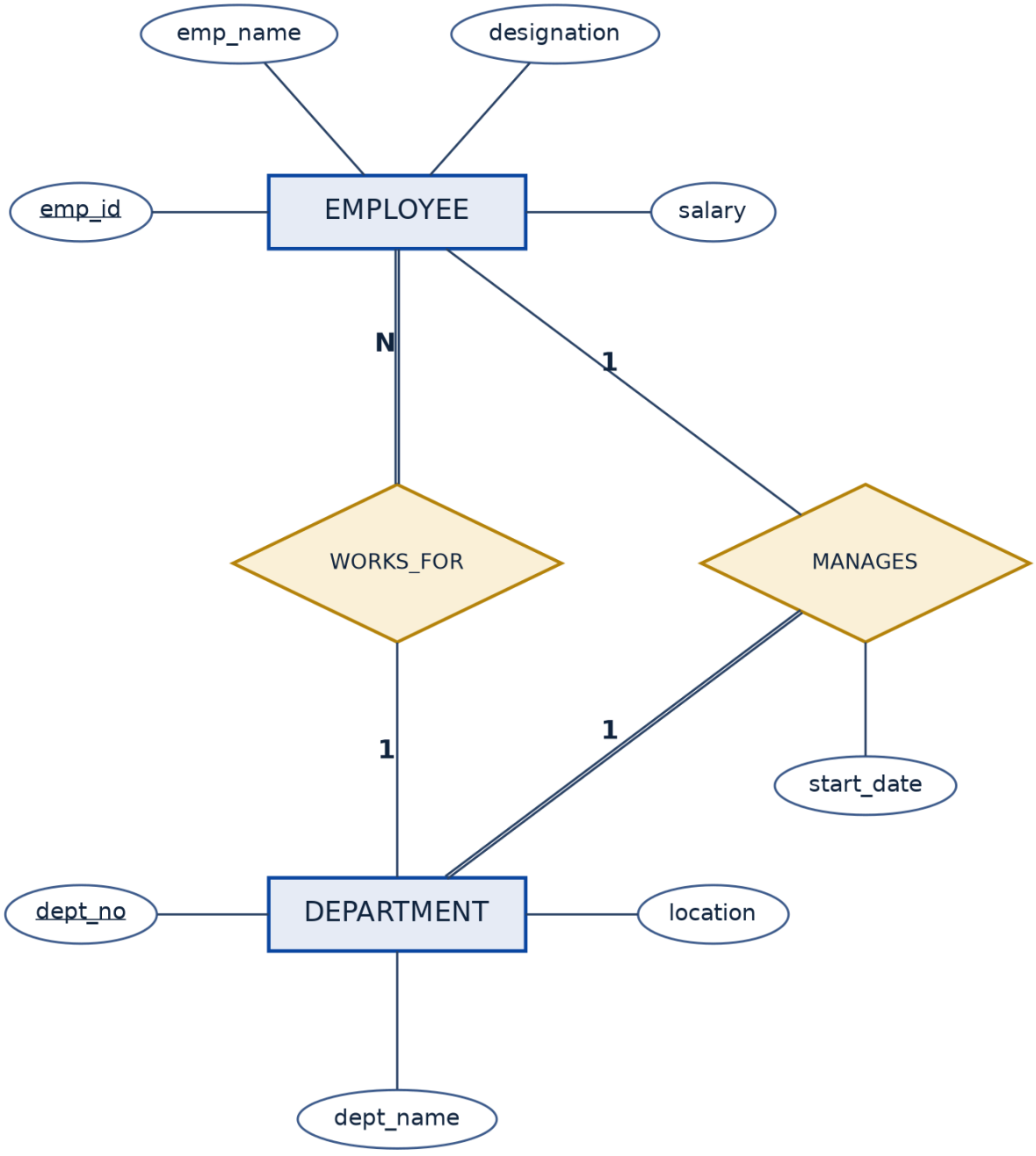


Figure 3 — ER diagram for Scenario 3: Company Employees and Departments

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Scenario 4 — Diagnostic Centre Appointments

Level: Easy – Intermediate **Concepts:** Composite attributes · multivalued attributes · derived attributes

The scenario

A diagnostic centre stores, for each patient, a patient ID, the patient's name (recorded as first name and last name separately), the date of birth, and a full address which is broken down into house number, road number and city. A patient may give any number of contact phone numbers. The patient's age must always appear on screen, but the centre has decided never to store it.

For each doctor the centre stores a doctor ID, name and consultation fee, together with the specialisations the doctor holds — a doctor may hold more than one specialisation.

A patient may take appointments with many doctors, and a doctor sees many patients. The date and the time of every appointment are recorded. Draw an ER diagram, paying particular attention to the type of each attribute.

Solution — entity sets and attributes

Entity set	Attributes (primary key underlined)
PATIENT	patient_id, name *(composite: first_name, last_name)*, date_of_birth, age *(derived)*, address *(composite: house_no, road_no, city)*, phone_no *(multivalued)*
DOCTOR	doctor_id, doctor_name, specialization *(multivalued)*, consultation_fee

Solution — relationship sets

Relationship	Between	Ratio	Participation	Attributes
APPOINTMENT	PATIENT – DOCTOR	M : N	Partial on both sides	appt_date, appt_time

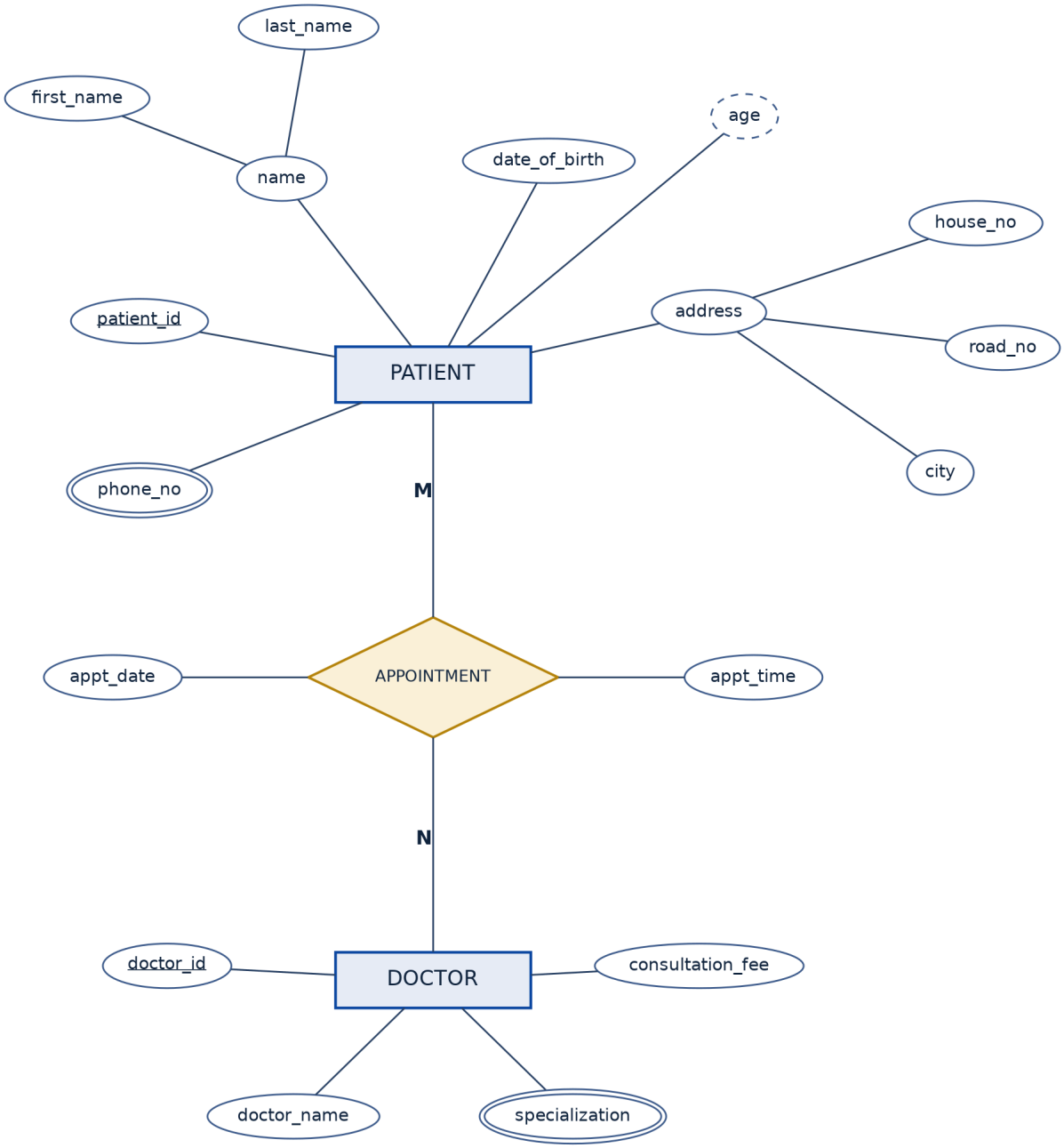


Figure 4 — ER diagram for Scenario 4: Diagnostic Centre Appointments

Scenario 5 — Bank Accounts and Transactions

Level: Intermediate **Concepts:** Weak entity · partial key · identifying relationship · total participation

The scenario

A bank stores, for every branch, a branch code, a branch name and a city. For every customer it stores a customer ID, name, address and phone number. For every account it stores an account number, the account type and the current balance.

Every account is maintained at exactly one branch. A customer may hold several accounts, and an account may be jointly held by several customers; every account must have at least one holder.

Every account has a history of transactions. Transactions are numbered 1, 2, 3 … within an account and the numbering starts again from 1 for the next account, so the same transaction number appears against many accounts. For each transaction the bank records the date, the amount and the type (deposit or withdrawal). If an account is closed and removed from the database, its entire transaction history disappears with it.

Draw an ER diagram.

Solution — entity sets and attributes

Entity set	Attributes (primary key underlined)
CUSTOMER	cust_id, cust_name, address, phone
ACCOUNT	acc_no, acc_type, balance
BRANCH	branch_code, branch_name, city
TRANSACTION (weak)	*trans_no* *(partial key)*, trans_date, amount, trans_type

Solution — relationship sets

Relationship	Between	Ratio	Participation	Attributes
HOLDS	CUSTOMER – ACCOUNT	M : N	Total on ACCOUNT, partial on CUSTOMER	—
MAINTAINS	ACCOUNT – BRANCH	N : 1	Total on ACCOUNT, partial on BRANCH	—
LOGS (identifying)	ACCOUNT – TRANSACTION	1 : N	Total on TRANSACTION	—

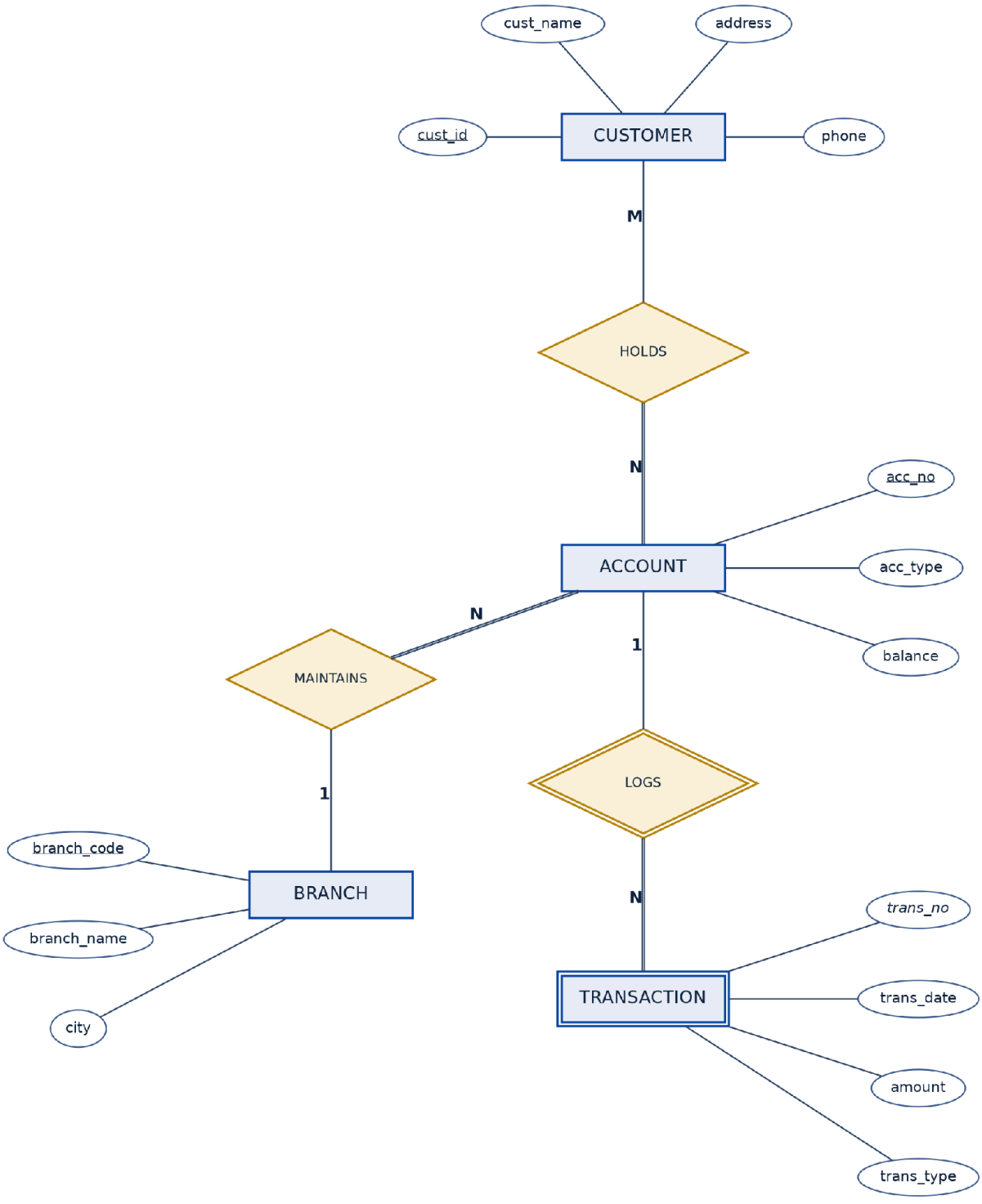


Figure 5 — ER diagram for Scenario 5: Bank Accounts and Transactions

Scenario 6 — Hotel Room Reservation

Level: Intermediate **Concepts:** Weak entity with a strong owner · M:N relationship on a weak entity

The scenario

A hotel chain stores, for each hotel, a hotel ID, name, city and star rating. Every hotel has many rooms. A room is identified by its room number, but room number 101 exists in almost every hotel of the chain, so the number is unique only within one hotel. For each room the chain stores the room type and the rate per night.

Guests are identified by a guest ID, and the chain stores the guest's name, national ID number and phone number. A guest may book many rooms over time, and a room is booked by many guests over time. Each booking records the check-in date, the check-out date and the total bill.

A guest record is created at the time of the first booking. Draw an ER diagram.

Solution — entity sets and attributes

Entity set	Attributes (primary key underlined)
HOTEL	hotel_id, hotel_name, city, star_rating
ROOM (weak)	*room_no (partial key)*, room_type, rate_per_night
GUEST	guest_id, guest_name, nid_no, phone

Solution — relationship sets

Relationship	Between	Ratio	Participation	Attributes
HAS (identifying)	HOTEL – ROOM	1 : N	Total on ROOM	—
BOOKS	ROOM – GUEST	M : N	Total on GUEST, partial on ROOM	check_in, check_out, total_bill

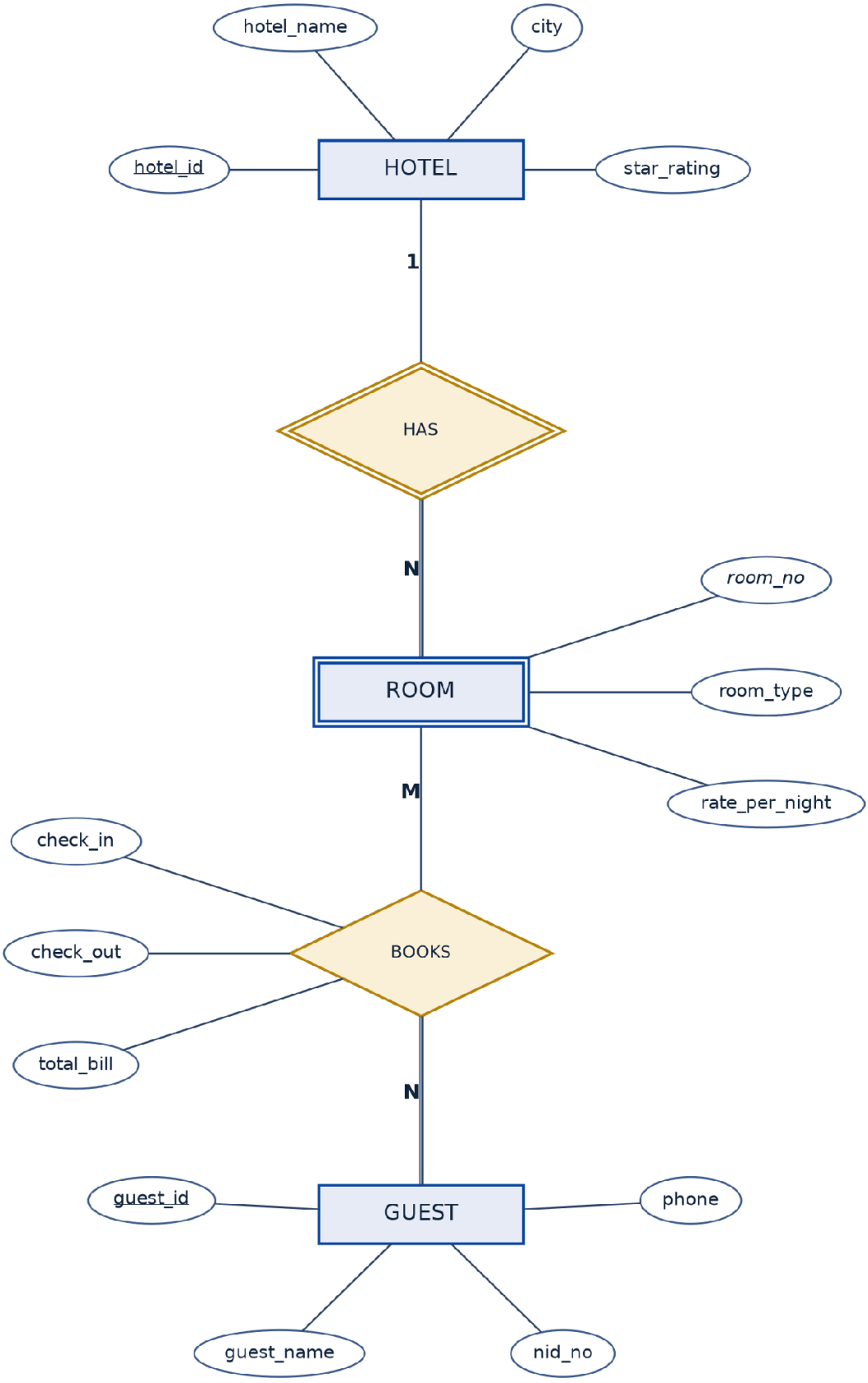

hotel_name
city
hotel_id
HOTEL
star_rating
1
HAS
N
room_no
ROOM
room_type
M
rate_per_night
check_in
check_out
BOOKS
total_bill
N
guest_id
GUEST
phone
guest_name
nid_no

Figure 6 — ER diagram for Scenario 6: Hotel Room Reservation

Scenario 7 — University Academic Management

Level: Intermediate **Concepts:** Generalisation / specialisation (disjoint, total) · weak entity · recursive M:N · multiple relationships

The scenario

A university keeps a single record for every person on campus, storing a person ID, full name, date of birth and address. Every such person is either a student or a teacher, never both, and nobody is stored who is neither. For a student the university additionally stores a roll number, CGPA and level–term. For a teacher it stores designation, joining date and salary.

Draw an ER diagram. (The solution is presented as two figures for legibility — they form one diagram, joined by STUDENT and TEACHER.)

Solution — entity sets and attributes

Entity set	Attributes (primary key underlined)
PERSON	person_id, full_name, date_of_birth, address
STUDENT	roll_no, cgpa, level_term
TEACHER	designation, joining_date, salary

Solution — relationship sets

Relationship	Between	Ratio	Participation	Attributes
ISA	PERSON ⟶ STUDENT / TEACHER	—	Total, disjoint (d)	—

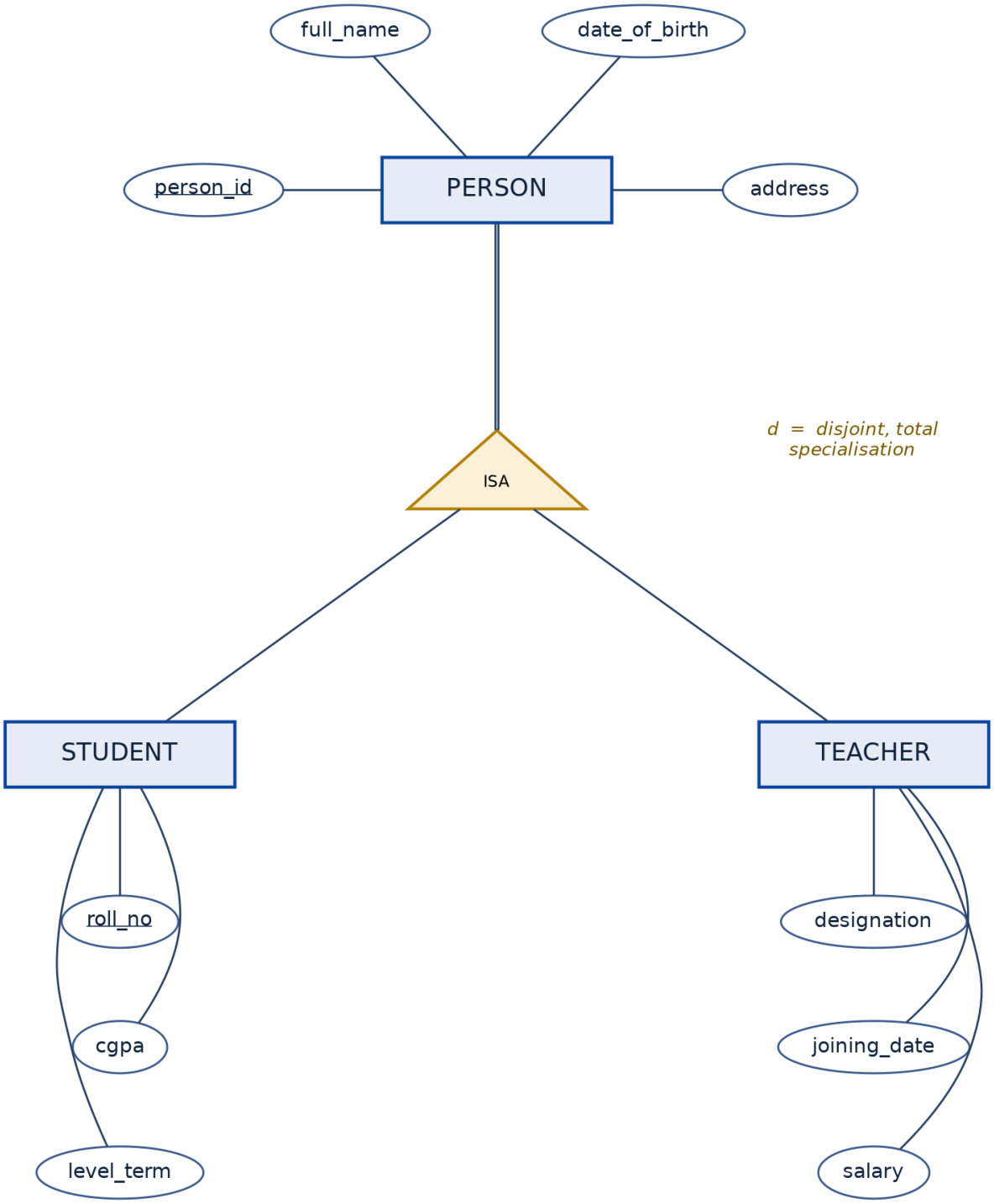
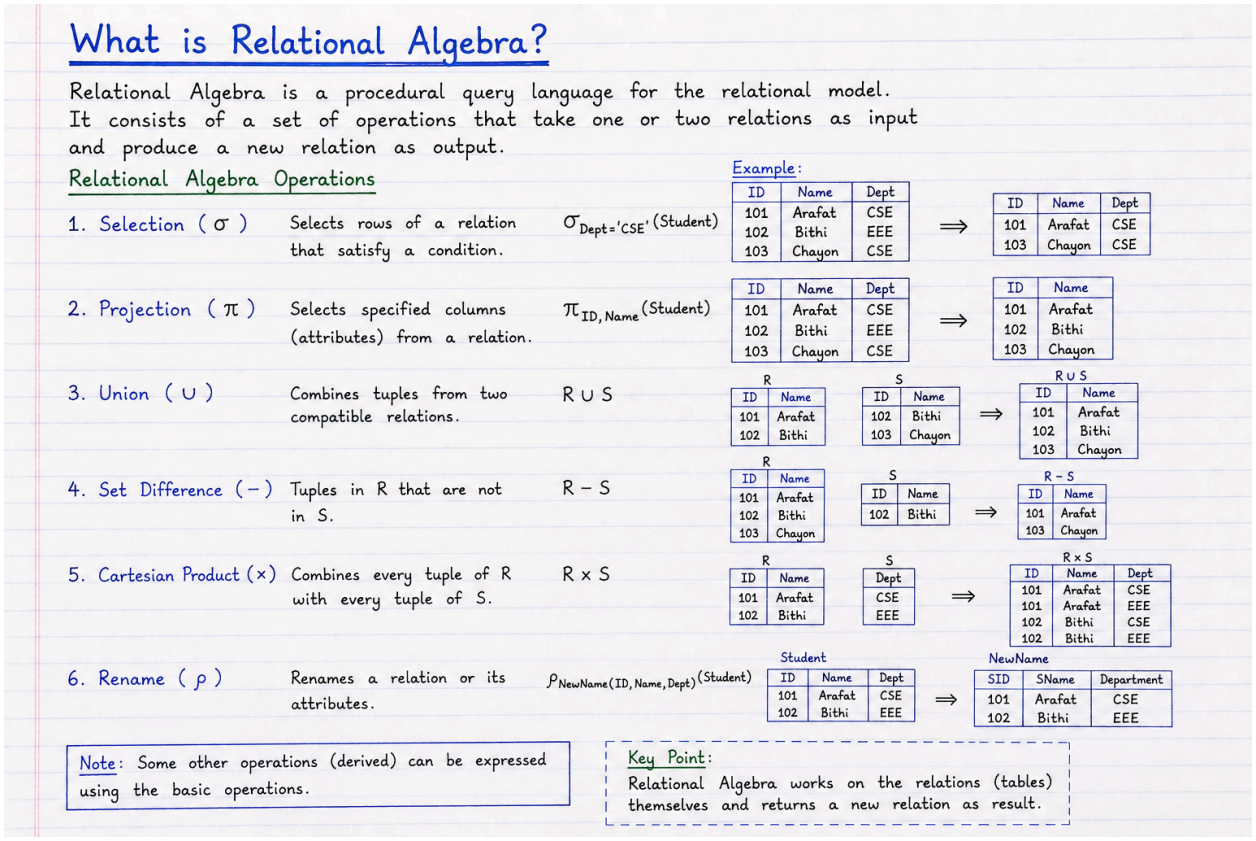


Figure 7— Generalisation / specialisation of PERSON

Week 3

What is Relational Algebra?



What is Relational Algebra?

Relational Algebra is a procedural query language for the relational model. It consists of a set of operations that take one or two relations as input and produce a new relation as output.

Relational Algebra Operations

1. **Selection (σ)** — Selects rows of a relation that satisfy a condition. $\sigma_{Dept='CSE'}(Student)$

Example:

ID	Name	Dept
101	Arafat	CSE
102	Bithi	EEE
103	Chayon	CSE

⟹

ID	Name	Dept
101	Arafat	CSE
103	Chayon	CSE

2. **Projection (π)** — Selects specified columns (attributes) from a relation. $\pi_{ID, Name}(Student)$

ID	Name	Dept
101	Arafat	CSE
102	Bithi	EEE
103	Chayon	CSE

⟹

ID	Name
101	Arafat
102	Bithi
103	Chayon

3. **Union (∪)** — Combines tuples from two compatible relations. $R \cup S$

R

ID	Name
101	Arafat
102	Bithi

S

ID	Name
102	Bithi
103	Chayon

⟹

R ∪ S

ID	Name
101	Arafat
102	Bithi
103	Chayon

4. **Set Difference (−)** — Tuples in R that are not in S. $R - S$

R

ID	Name
101	Arafat
102	Bithi
103	Chayon

S

ID	Name
102	Bithi

⟹

R − S

ID	Name
101	Arafat
103	Chayon

5. **Cartesian Product (×)** — Combines every tuple of R with every tuple of S. $R \times S$

R

ID	Name
101	Arafat
102	Bithi

S

Dept
CSE
EEE

⟹

R × S

ID	Name	Dept
101	Arafat	CSE
101	Arafat	EEE
102	Bithi	CSE
102	Bithi	EEE

6. **Rename (ρ)** — Renames a relation or its attributes. $\rho_{NewName(ID, Name, Dept)}(Student)$

Student

ID	Name	Dept
101	Arafat	CSE
102	Bithi	EEE

⟹

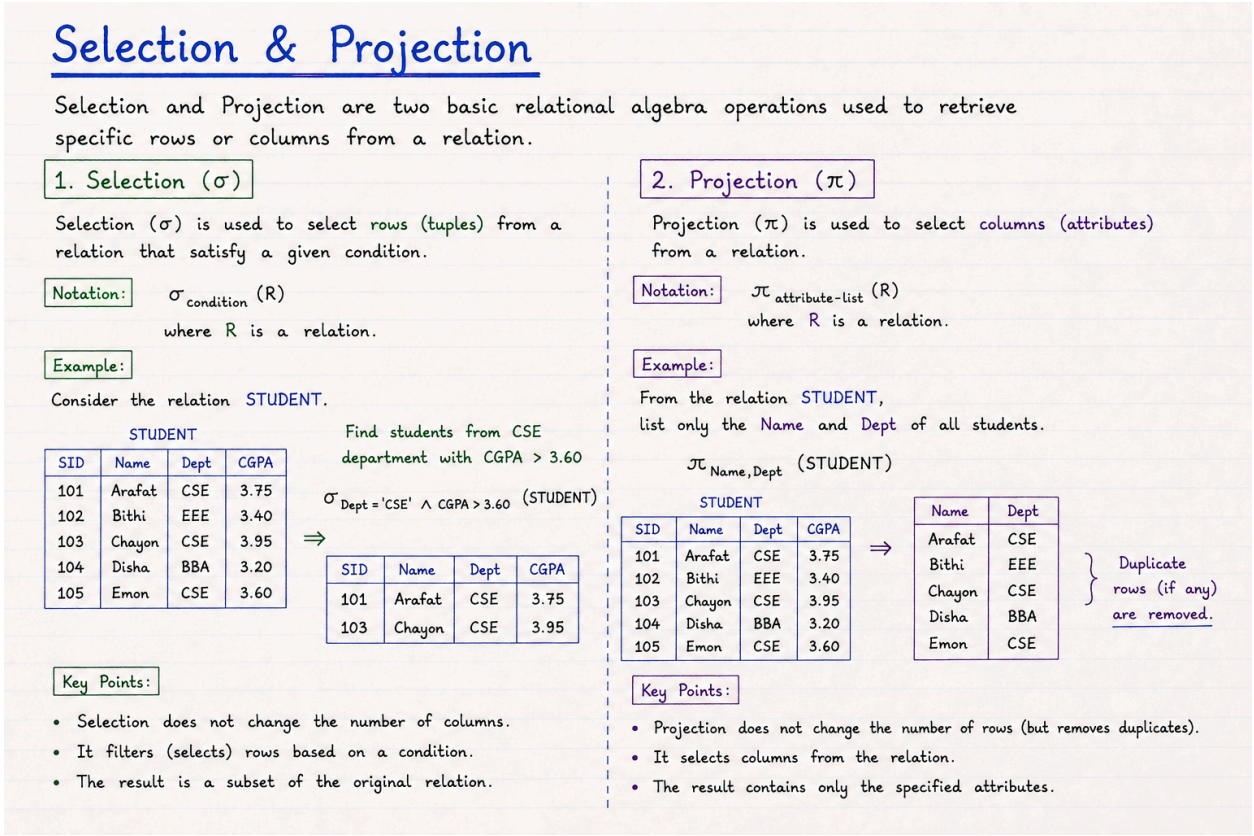
NewName

SID	SName	Department
101	Arafat	CSE
102	Bithi	EEE

Note: Some other operations (derived) can be expressed using the basic operations.

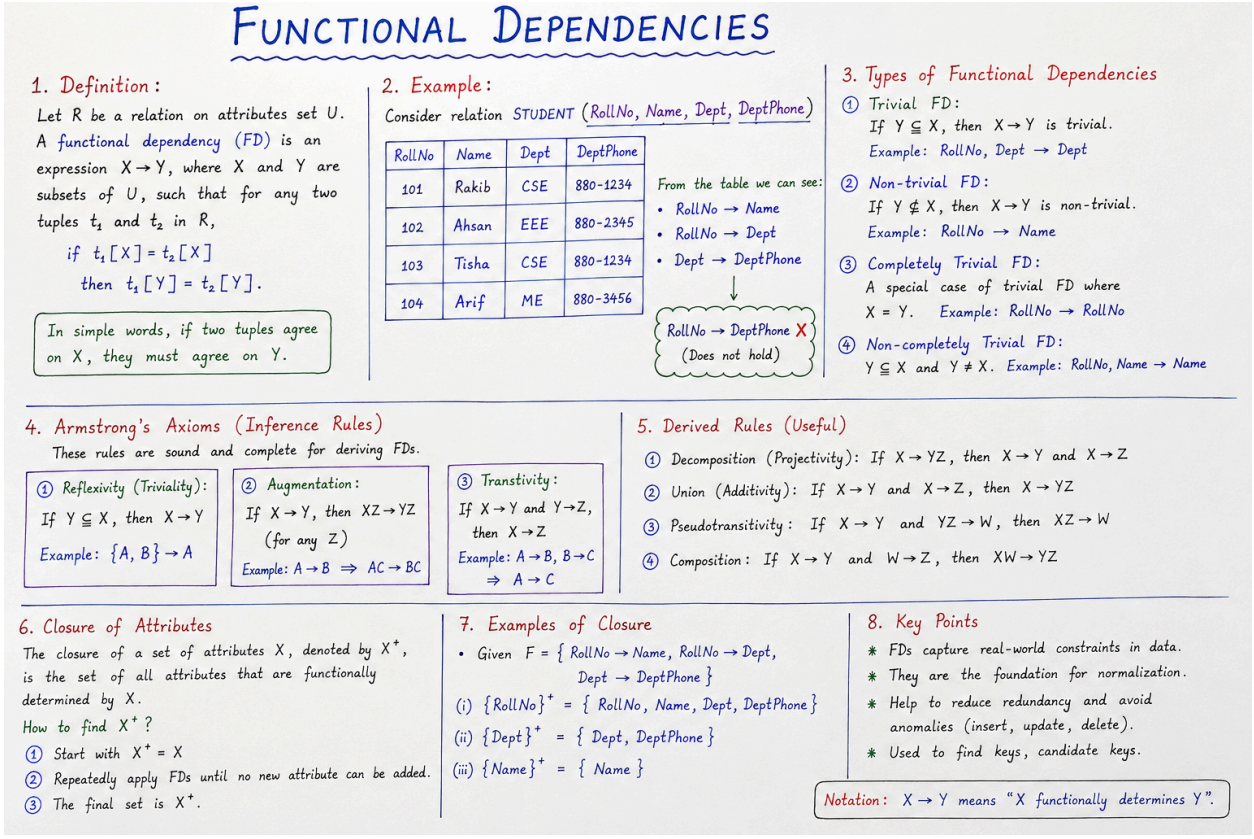
Key Point:
Relational Algebra works on the relations (tables) themselves and returns a new relation as result.

Selection & Projection



Week 4

Functional Dependencies



Normalization

Normalization

Definition : Normalization is the process of organizing data in a database to reduce redundancy and improve data integrity by dividing large tables into smaller, related tables and defining relationships between them.

Goals of Normalization :

1) Minimize data redundancy.
2) Eliminate update anomalies (insert, delete, update anomalies).
3) Ensure data integrity and consistency.
4) Improve storage efficiency.

Normal Forms in Normalization :

1NF (First Normal Form)

- All attributes are atomic (indivisible).
- Each row is unique.
- No repeating groups.

Example : Student

StudentID	StudentName	Course	PhoneNumber
101	Rakib	DBMS	01711-111111
101	Rakib	OS	01711-111111
101	Rakib	CN	01711-111111
102	Sadia	DBMS	01622-222222
103	Tanmoy	OS	01833-333333

Key Point :
In 1NF, all values are atomic and there is no repeating group.

2NF (Second Normal Form)

- Must be in 1NF.
- All non-key attributes are fully functionally dependent on the whole of the primary key.
- No partial dependency.

Example :

Table 1 : Student

StudentID	PhoneNumber
101	01711-111111
102	01622-222222
103	01833-333333

Table 2 : StudentCourse

StudentID	Course
101	DBMS
101	OS
101	CN
102	DBMS
103	OS

Key Point :
All non-key attributes are fully dependent on the whole primary key.

3NF (Third Normal Form)

- Must be in 2NF.
- No transitive dependency.
- Non-key attributes are not dependent on other non-key attributes.

Example :

Table 1 : Student

StudentID	PhoneNumber
101	01711-111111
102	01622-222222
103	01833-333333

Table 2 : StudentCourse

StudentID	CourseID
101	C01
101	C02
101	C03
102	C01
103	C02

Table 3 : Course

CourseID	CourseName
C01	Database Management System
C02	Operating System
C03	Computer Network

Key Point :
No transitive dependency.
Every non-key attribute depends only on the primary key.

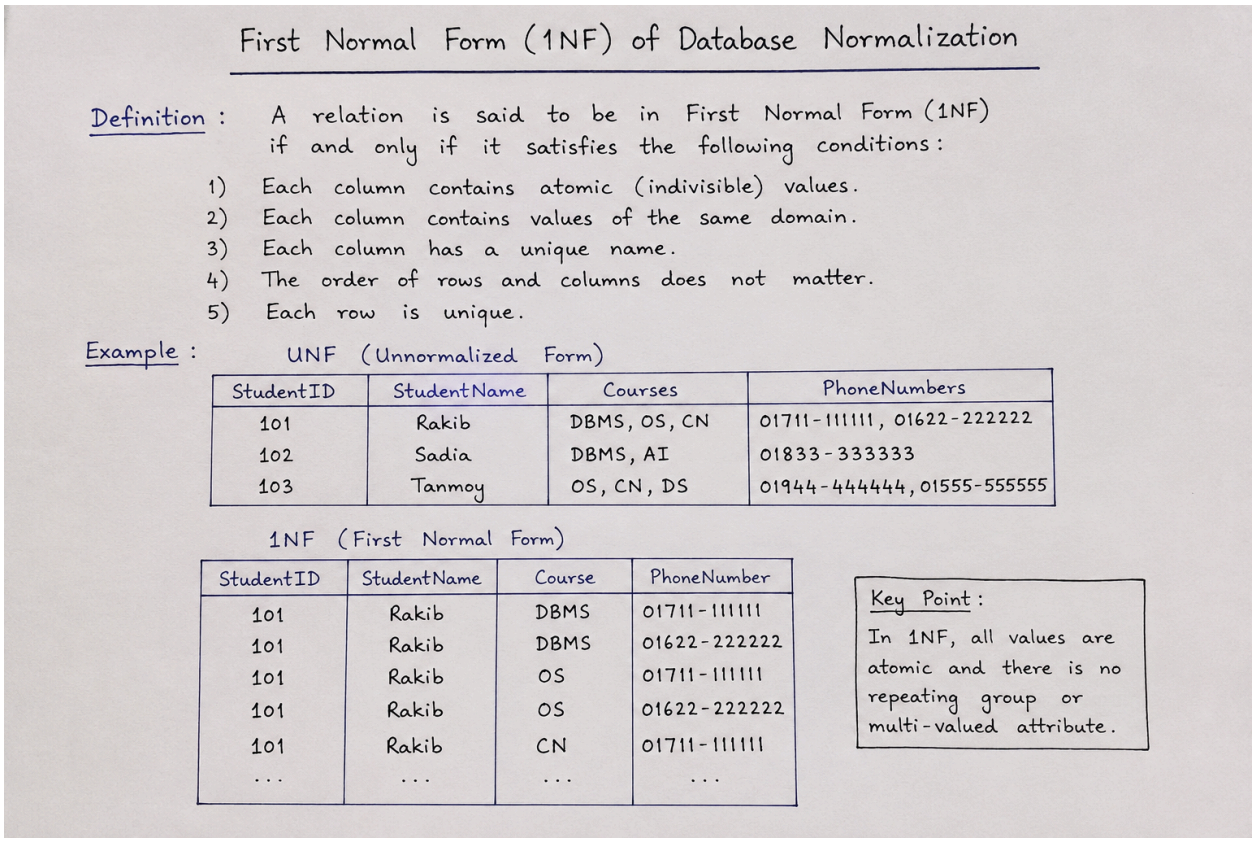
Summary of Normalization

Normal Form	Main Rule
1NF	Atomic values, no repeating groups, unique rows.
2NF	In 1NF and no partial dependency.
3NF	In 2NF and no transitive dependency.

Benefits :

- Reduced data redundancy
- Prevents anomalies
- Better data integrity
- Easier maintenance
- Efficient storage

First Normal Form (1NF)



First Normal Form (1NF) of Database Normalization

Definition : A relation is said to be in First Normal Form (1NF) if and only if it satisfies the following conditions :

1) Each column contains atomic (indivisible) values.
2) Each column contains values of the same domain.
3) Each column has a unique name.
4) The order of rows and columns does not matter.
5) Each row is unique.

Example :

UNF (Unnormalized Form)

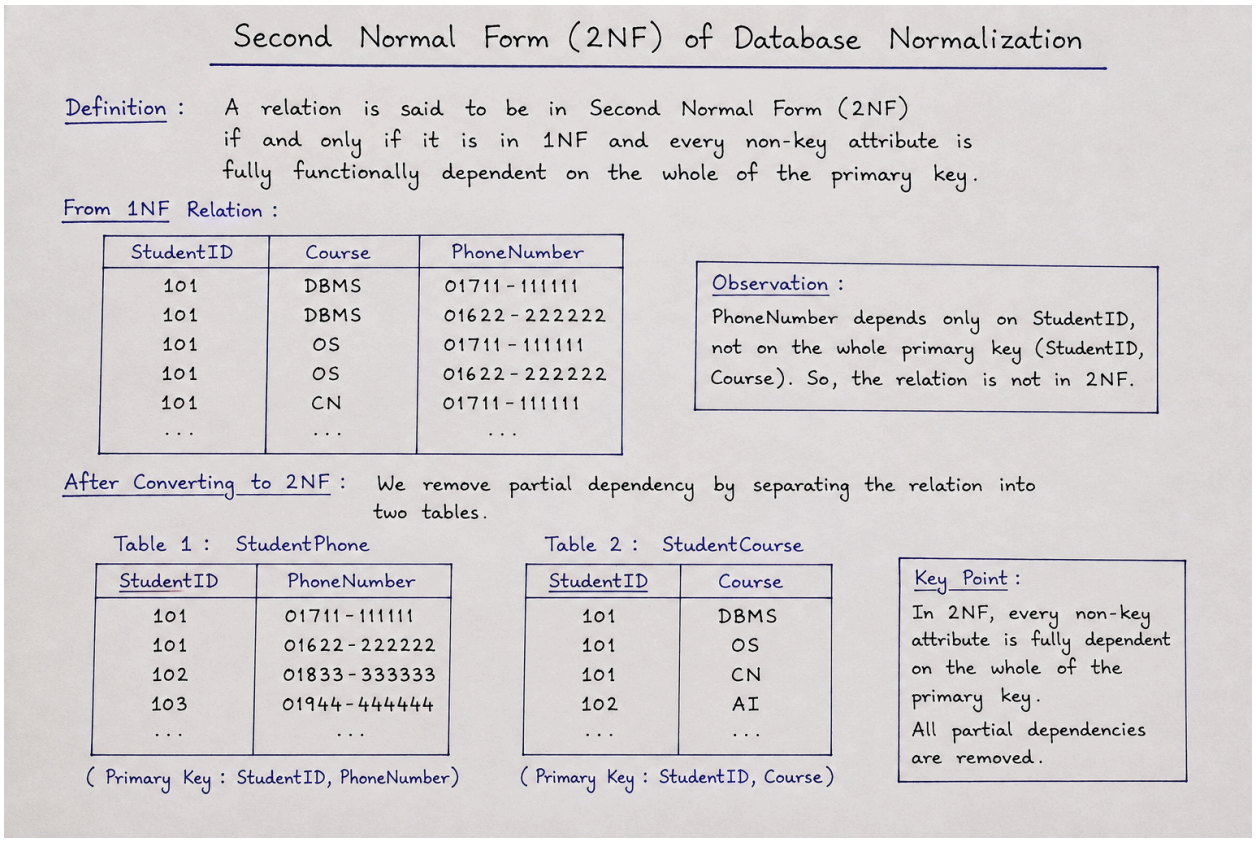
StudentID	StudentName	Courses	PhoneNumbers
101	Rakib	DBMS, OS, CN	01711-111111, 01622-222222
102	Sadia	DBMS, AI	01833-333333
103	Tanmoy	OS, CN, DS	01944-444444, 01555-555555

1NF (First Normal Form)

StudentID	StudentName	Course	PhoneNumber
101	Rakib	DBMS	01711-111111
101	Rakib	DBMS	01622-222222
101	Rakib	OS	01711-111111
101	Rakib	OS	01622-222222
101	Rakib	CN	01711-111111
...	...	...	...

Key Point :
In 1NF, all values are atomic and there is no repeating group or multi-valued attribute.

Second Normal Form (2NF)



Second Normal Form (2NF) of Database Normalization

Definition : A relation is said to be in Second Normal Form (2NF) if and only if it is in 1NF and every non-key attribute is fully functionally dependent on the whole of the primary key.

From 1NF Relation :

StudentID	Course	PhoneNumber
101	DBMS	01711-111111
101	DBMS	01622-222222
101	OS	01711-111111
101	OS	01622-222222
101	CN	01711-111111
...	...	...

Observation :
PhoneNumber depends only on StudentID, not on the whole primary key (StudentID, Course). So, the relation is not in 2NF.

After Converting to 2NF : We remove partial dependency by separating the relation into two tables.

Table 1 : StudentPhone

StudentID	PhoneNumber
101	01711-111111
101	01622-222222
102	01833-333333
103	01944-444444
...	...

(Primary Key : StudentID, PhoneNumber)

Table 2 : StudentCourse

StudentID	Course
101	DBMS
101	OS
101	CN
102	AI
...	...

(Primary Key : StudentID, Course)

Key Point :
In 2NF, every non-key attribute is fully dependent on the whole of the primary key.
All partial dependencies are removed.

Third Normal Form (3NF)